norms

$Q^* = Q^{-1} \Leftrightarrow QQ^* = Q^*Q = I \Leftrightarrow$ cols orthonormal

unitary preserve: $(Qx)^*(Qy) = x^*y$

$\|Qx\| = \|x\|$

Cauchy Sw: $\langle x,y\rangle \le \|x\|\|y\|$

$\|A\|_{(m,n)} := \sup_{\|x\|_n=1} \|Ax\|_m$

$\|AB\|_{(l,n)} \le \|A\|_{l,m}\|B\|_{m,n}$

$\|A\|_1 = \max_{1\le j\le n} \|a_j\|_1$ (= max abs col sum), $\|A\|_2 = \sigma_1$

$\|A\|_\infty$ = max abs row sum

$\|A\|_F = \sqrt{\sum\sum |a_{ij}|^2} = \sqrt{\sum_n \|\vec{a_j}\|^2} = \sqrt{tr(A^*A)} = \sqrt{tr(AA^*)} = \sqrt{\sum_1^r \sigma_i^2}$

$\|AB\|_F^2 \le \|A\|_F^2\|B\|_F^2$, $\|QA\|_2 = \|A\|_2$, $\|QA\|_F = \|A\|_F$

projector

projector $P: P^2 = P$ ($\Rightarrow P^n = P$) (not ortho) (ortho)

P projector $\Rightarrow I-P$ also projector

$V \xrightarrow[S_1]{\text{along}} S_2$ $\quad V \xrightarrow[S_2]{\text{along}} S_1$

Pv, $im(P)$, Pv, $im(P)$

$(Pv - v) \in null(P)$

P orthogonal projector $\Leftrightarrow S_1 \perp S_2 \Leftrightarrow P = P^*$

P orthogonal projector $\Rightarrow I-P$ also orthogonal projector

orthogonal proj $P \Rightarrow$ $\boxed{P = Q\Sigma Q^*}$ both SVD and eigendecomp

$\{q_1, \dots, q_n\}$ span S_1, $\{q_{n+1}, \dots, q_m\}$ span S_2

$\Sigma: \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 0 & \\ & & & 0 \end{bmatrix}$

can just take $[q_1, \dots, q_n]$ as $\hat{Q} \Rightarrow \boxed{P = \hat{Q}\hat{Q}^*}$ (orthogonal proj onto $im(\hat{Q})$)

specially, proj onto line of $\langle a\rangle$: $P_{\langle a\rangle} = \frac{aa^*}{a^*a}$

Given arbitrary basis $\{a_1, \dots, a_n\}$, $P_{\langle a_1,\dots,a_n\rangle} = A(A^*A)^{-1}A^*$

SVD

SVD: $v_i \mapsto \sigma_i u_i$

reduced: $A = \hat{U}\hat{\Sigma}V^*$
$\quad m\times n$, $n\times n$, $n\times n$; (ortho cols) diag (ortho cols)

full: $A = U\Sigma V^*$
$\quad m\times m$, $m\times n$, $n\times n$; uni diag uni

solve $Ax = b$ by SVD: $A = U\Sigma V^*$, $x' := V^*x$, $b' := U^*b \Rightarrow b' = \Sigma x'$

1. $rank(A) = \langle u_1, \dots, u_r\rangle$, $null(A) = \langle v_{r+1}, \dots, v_n\rangle$
2. nonzero $\sigma(A)$ is $\sqrt{\lambda(A^*A)}$
3. $|\det A| = \prod \sigma_i$
4. $A = \sum_{j=1}^r \sigma_j u_j v_j^*$

$\|A - A_v\|_2 = \sigma_{v+1}$, $\quad \sum^v \sigma_j u_j v_j^*$

$\|A - A_v\|_F = \sqrt{\sigma_{v+1}^2 + \dots + \sigma_n^2}$

SVD 1, SVD 2

计算 SVD:

1. 求解 A^*A (or AA^*) 的 eigenvalues, $\sigma_i = \sqrt{\lambda_i}$
2. A^*A 的 eigenvectors (unit), $V = [\vec{v_1} \dots \vec{v_n}]$ (可能需要补全)
3. $Av_i = \sigma_i u_i$, 求解 $U = [\vec{u_1}, \dots, \vec{u_n}]$
4. $A = U\Sigma V^*$

eigenvalue decomp: $A \in \mathbb{C}^{m\times m}$, $A = X\Lambda X^{-1}$, X 为 eigenvectors
A diagonalizable ($GM = AM$) $\quad (A - \lambda I)x = 0$

QR

QR $\quad A = QR$ ($m\times m$, $m\times n$) $= \hat{Q}\hat{R}$ ($m\times n$, $n\times n$) $\quad \vec{a_n} = \sum_{j=1}^n r_{jn}\hat{q_n}$

projector

QR def

Classical GS:

$v_j -= \sum_{i=1}^{j-1} q_i q_i^* a_j$

i.e. $v_j = (I - \hat{Q}_{j-1}\hat{Q}_{j-1}^*) a_j$ $:= P_j$

$P_1 = I$

$O(n)$ for $j = 1$ to n
$\quad v_j = a_j$
$O(mn)$ $\quad$ for $i = 1$ to $j-1$
$\quad\quad r_{ij} = q_i^* a_j$
$\quad\quad v_j -= r_{ij} q_i$ $\quad$ ($-\text{proj}_{q_i} a_j$)
$\quad r_{jj} = \|v_j\|_2$
$\quad q_j = v_j / r_{jj}$ (normalize)

$v_j -= \text{proj}_{\langle q_1, \dots, q_{j-1}\rangle} v_j$

$q_j = \frac{v_j}{\|v_j\|}$

Thm: $A \in \mathbb{C}^{m\times n}$ full rank $\Rightarrow$ unique reduced QR

ex. (classical GS)

$A = \begin{bmatrix} 1 & -1 \\ 1 & 5 \\ 0 & 6 \end{bmatrix}$, $v_2 -= (q_1^* a_2) q_1$

$\begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ -0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{4}{\sqrt{2}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\sqrt{54}} \end{bmatrix}$

normalize v_1: $q_1 = \frac{v_1}{\|v_1\|} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{bmatrix}$; normalize v_2

$= \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{3}{\sqrt{6}} \\ \frac{\sqrt{2}}{2} & \frac{3}{\sqrt{6}} \\ 0 & \frac{6}{\sqrt{54}} \end{bmatrix} = \hat{Q}$

$\hat{R} = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{54} \end{bmatrix} \begin{bmatrix} 1 & \frac{4}{\sqrt{2}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 0 & 1 \end{bmatrix}$

Householder

for $k = 1$ to n $\quad 2mn^2 - \frac{2}{3}n^3$
$\quad x = A_{k:m,k}$
$\quad v_k = sign(x_1)\|x\|_2 e_1 + x$
$\quad v_k = \frac{v_k}{\|v_k\|_2}$
$\quad A_{k:m,k:n} = A_{k:m,k:n} - 2v_k(v_k^* A_{k:m,k:n})$

implicit calculate Q^*b
for $k = 1$ to n
$\quad b_{k:m} = b_{k:m} - 2v_k(v_k^* b_{k:m})$

implicit calculate Qx
for $k = n$ to 1
$\quad x_{k:m} = x_{k:m} - 2v_k(v_k^* x_{k:m})$

$Q_k = \begin{bmatrix} I_{k-1} & \\ & F \end{bmatrix}$

$F = I - 2\frac{vv^*}{\|v\|}$

即 Q_k 把 span(x) $=$ span($A_{k:m,k}$) reflect 到 span(e_1) 上

$\underbrace{Q_n \dots Q_2 Q_1}_{= Q^*} A = R$

Modified GS:

for $i = 1$ to n
$\quad v_i = a_i$
for $i = 1$ to n
$\quad r_{ii} = \|v_i\|$
$\quad q_i = v_i / r_{ii}$
$\quad$ for $j = i+1$ to n
$\quad\quad r_{ij} = q_i^* v_j$
$\quad\quad v_j = v_j - r_{ij} q_i$

work: $\sim 2mn^2$ flops

matrix form: $\underbrace{A R_1 R_2 \dots R_n}_{\hat{R}^{-1}} = \hat{Q}$

$R_1 = \begin{bmatrix} \frac{1}{r_{11}} & -\frac{r_{12}}{r_{11}} & -\frac{r_{13}}{r_{11}} & \cdots \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$

$R_2 = \begin{bmatrix} 1 & & & \\ & \frac{1}{r_{22}} & -\frac{r_{23}}{r_{22}} & \cdots \\ & & 1 & \\ & & & 1 \end{bmatrix}$

To get full QR: find $b_3 \perp im(A)$, normalize it

$Q = \begin{bmatrix} \hat{Q} & \frac{b_3}{\|b_3\|} \end{bmatrix}$ $\quad \hat{R} = \begin{bmatrix} R \\ 0 \end{bmatrix}$

Least squares

$\min\|r\|_2 \Leftrightarrow A^*r = 0 \Leftrightarrow A^*Ax = A^*b \Leftrightarrow \text{Proj}_A b = Ax$

use SVD to solve least sq. $\quad A^+ = (A^*A)^{-1}A^*$

$A = \hat{U}\hat{\Sigma}V^*$ $\quad$ note: $im(A) = im(\hat{U})$, and U has orthonormal cols

$2mn^2 + 11n^3$

$\Rightarrow Ax_{op} = UU^*b$ (or $\hat{U}\hat{U}^*$)

$x_{op} = A^+b = V\Sigma^+U^*b$ $\quad$ pesudoinverse 取 $\frac{1}{\sigma_i}$ 即可

$2mn^2 - \frac{2}{3}n^3$

use QR of A to solve least sq:

$\text{proj}_A b = \hat{Q}\hat{Q}^*b = Ax \Rightarrow \hat{R}x = \hat{Q}^*b \Rightarrow x = \hat{R}^{-1}\hat{Q}^*b$

FFT

$\tilde{F}_{2n} = \begin{bmatrix} F_n & T_nF_n \\ F_n & -T_nF_n \end{bmatrix}$ $\quad w_{2n} = e^{\frac{-2\pi i}{2n}}$

$T_n = diag(1, w_{2n}^1, w_{2n}^2, \dots, w_{2n}^{n-1})$

F_{2n} = 重排 $\tilde{F}_{2n}$ 的列, by 二进制反转

ex $\quad 0 = 00_2 \to 0 = 00_2$, $\quad 1 = 01_2 \to 2 = 10_2$, $\quad 2 = 10_2 \to 1 = 01_2$, $\quad 3 = 11_2 \to 3 = 11_2$

ex $\tilde{F}_4 = \begin{bmatrix} F_2 & \begin{bmatrix} 1 & \\ & -i \end{bmatrix} F_2 \\ F_2 & -\begin{bmatrix} 1 & \\ & -i \end{bmatrix} F_2 \end{bmatrix}$

S_n eigenvalue: $\{\lambda_k\}_{k=0}^{n-1}$, $\lambda_k = e^{\frac{2\pi i k}{n}}$

$\Rightarrow F_n^* S_n F_n = D_n = \begin{bmatrix} w^0 & & & \\ & w^1 & & \\ & & \ddots & \\ & & & w^{n-1} \end{bmatrix}$

$F_n^*(P(S_n))F_n = (P(D))$

for $p(z) = \sum_{j=0}^{n-1} c_j z^j = c_0 + c_1 z + c_2 z^2 + \dots + c_{n-1} z^{n-1}$

Define: $P_{even}(z) := c_0 + c_2 z + \dots + c_{n-2} z^{\frac{n}{2}-1}$
$\quad P_{odd}(z) := c_1 + c_3 z + \dots + c_{n-1} z^{\frac{n}{2}-1}$

$\Rightarrow p(z) = P_{even}(z^2) + z P_{odd}(z^2)$

$= \frac{(1+z^{\frac{n}{2}})}{2} P_{even}(z) + \frac{(1-z^{\frac{n}{2}})}{2} P_{odd}(z/w_n)$

ex $n = 4$, $p(z) = a + bz + cz^2 + dz^3$
Given $p(1) = 2$, $p(i) = 3$, $p(-1) = 5$, $p(-i) = 8$ (ω)
求 a, b, c, d

Sol $\quad p(z) = \frac{1+z^2}{2} P_{even}(z) + \frac{1-z^2}{2} P_{odd}(z/i)$ $\quad$ ($a + cz$, $b + dz$)

note, $P_{even}(z)$ 过 $(1,2)$ 和 $(-1,5)$ $\Rightarrow P_{even}(z) = \frac{1+z}{2} \cdot 2 + \frac{1-z}{2} \cdot 5 = \frac{7-3z}{2}$

同理, $P_{odd}(z) = \frac{1+z}{2} \cdot 3 + \frac{1-z}{2} \cdot 8 = \frac{11-5z}{2}$

$\Rightarrow p(z) = \frac{1+z^2}{2} \cdot \frac{7-3z}{2} + \frac{1-z^2}{2} \cdot \frac{11-5(z/i)}{2}$